

The gap is bounded by the variance of the averaged spatial Polyakov loop, with the small-volume scaling built in

For the Kogut–Susskind Hamiltonian on the periodic lattice of side $L = N_s a$, the Feynman–Bijl inequality with the transverse average of the spatial Polyakov loop in direction i ,

$$\bar{P}_i = \frac{1}{N_s^2} \sum_{\text{lines } \parallel i} \text{Re tr} (U_{\ell_1} \cdots U_{\ell_{N_s}}),$$

gives the exact bound

$$\Delta_{a,L}^{\text{phys}} \leq \frac{\hbar c g^2}{2a} \frac{\langle \sum_{\ell} |\nabla_{\ell} \bar{P}_i|^2 \rangle_0}{\text{Var}_0 \bar{P}_i} \leq \frac{\hbar c g^2 N}{4L \text{Var}_0 \bar{P}_i},$$

for every coupling and lattice size. Three consequences. In a small box at weak coupling, where the ground state is the constant-mode state of C133 at leading order, numerator and denominator are both computable there and the bound is $(\hbar c/L) g^{2/3}$ times a pure number, the scaling of the exact zero-mode gap, so the upper side of the small-volume corner S of the obligations map follows from an exact inequality plus the zero-mode expectation values. In the infinite-volume limit, a mass gap m forces the averaged spatial Polyakov loop to self-average at least like $1/L$: $\text{Var}_0 \bar{P}_i \leq \hbar c g^2 N / (4mc^2 L)$, equivalently a bound on the transverse Polyakov-loop susceptibility, $\sum_y a^2 \langle \text{Re } P(x) \text{Re } P(x+y) \rangle_c \leq \hbar c g^2 N L / (4mc^2)$, which is a necessary condition for T3 expressed through one gauge-invariant correlation. At strong coupling the averaged loop has variance $1/(2N_s^2)$ and the bound is empty, so the plaquette bound of the upper-bound note and the present one cover complementary ends of the coupling axis. Nothing here is promoted.

1. The operator and the exact bound

Fix a direction i . The lattice decomposes into N_s^2 lines parallel to i , each of N_s links closing around the torus. For a line λ let $P_{\lambda} = \text{Re tr} \prod_{\ell \in \lambda} U_{\ell}$, the ordered product along the line; it is gauge invariant (a closed loop), and $\bar{P}_i = N_s^{-2} \sum_{\lambda} P_{\lambda}$. Let $O = \bar{P}_i - \langle \bar{P}_i \rangle_0$.

Theorem 1. For every $g > 0$, $N_s \geq 2$ and compact G with the normalization of the obligations map,

$$\Delta_{a,L}^{\text{phys}} \leq \frac{\hbar c g^2}{2a} \frac{\langle \sum_{\ell} |\nabla_{\ell} \bar{P}_i|^2 \rangle_0}{\text{Var}_0 \bar{P}_i} \leq \frac{\hbar c g^2 N}{4L \text{Var}_0 \bar{P}_i}.$$

Proof. O is a real gauge-invariant function of the links with zero mean, so Proposition 1 of the upper-bound note applies and gives the first inequality. Each link in direction i belongs to exactly one line and links in other directions do not enter $\bar{P}_i$, so $\nabla_{\ell} \bar{P}_i = N_s^{-2} \nabla_{\ell} P_{\lambda(\ell)}$. For a link of λ , $X_a \text{tr} \prod U = \text{tr}(T^a M)$ up to a phase with M a cyclic rearrangement of the product, and the completeness relation gives $\sum_a |\text{tr}(T^a M)|^2 = \frac{1}{2}(N - |\text{tr } M|^2/N) \leq N/2$; the real part has smaller gradient. Hence $\sum_{\ell} |\nabla_{\ell} \bar{P}_i|^2 \leq N_s^{-4} \cdot N_s^2 \cdot N_s \cdot N/2 = N/(2N_s)$, and $\frac{\hbar c g^2}{2a} \cdot \frac{N}{2N_s} = \frac{\hbar c g^2 N}{4L}$. $\square$

2. Weak coupling in a small box

At weak coupling the links are near the identity and the gauge-invariant content of a Polyakov line is the holonomy of the constant mode: $\prod_{\ell \in \lambda} U_{\ell} \approx \exp(iL a_i)$ with $a_i = a_i^b T^b$ the constant mode of G07 Proposition 7, whose ground-state width is $|a_i| \sim g^{2/3}/L$ in natural units. Since $\text{tr } T^b = 0$,

$$P_{\lambda} \approx \text{Re tr } e^{iL a_i} = N - \frac{1}{4} L^2 |\vec{a}_i|^2 + O(L^4 |a|^4), \quad \nabla_{\ell} P_{\lambda} \approx \text{Im tr}(T^a L a_i) = \frac{1}{2} L a_i^a,$$

so the variance of $\bar{P}_i$ is second order and the gradient first order in the constant mode:

$$\text{Var}_0 \bar{P}_i \approx \frac{1}{16} L^4 \text{Var}(|\vec{a}_i|^2), \quad \left\langle \sum_{\ell} |\nabla_{\ell} \bar{P}_i|^2 \right\rangle_0 \approx N_s^{-4} \cdot N_s^3 \cdot \frac{1}{4} L^2 \langle |\vec{a}_i|^2 \rangle = \frac{L^2 \langle |\vec{a}_i|^2 \rangle}{4N_s}.$$

In the dimensionless variables of C133, $La_i = g^{2/3}\xi_i$ with ξ_i distributed by the unit ground state $\psi_0^{(3)}$ of $\frac{1}{2}(-\Delta_{\mathbb{R}^9} + \sum_{i<j} |\vec{\xi}_i \times \vec{\xi}_j|^2)$:

$$\frac{\hbar c g^2}{2a} \cdot \frac{g^{4/3} \langle |\vec{\xi}_i|^2 \rangle / (4N_s)}{g^{8/3} \text{Var}(|\vec{\xi}_i|^2)/16} = \frac{\hbar c}{L} g^{2/3} \cdot \frac{2 \langle |\vec{\xi}_i|^2 \rangle}{\text{Var}(|\vec{\xi}_i|^2)}.$$

The bound is therefore $(\hbar c/L)g^{2/3}$ times the pure number $2 \langle |\vec{\xi}_i|^2 \rangle / \text{Var}(|\vec{\xi}_i|^2)$, finite because the unit ground state decays faster than any power along the valleys (the confining bound of C133 gives exponential decay of ψ_0 in $\sum_i |\vec{\xi}_i|$). Consistency with the exact zero-mode gap $\delta_1^{(3)} g^{2/3} \hbar c/L$ requires this number to be at least $\delta_1^{(3)}$, which is the Feynman–Bijl inequality applied inside the zero-mode model with the trial operator $|\vec{\xi}_i|^2$; the model version of Theorem 1 is exact and says exactly that.

What is proved and what is not: Theorem 1 is exact. The two approximations, replacing the ground-state expectations by their constant-mode values, are the leading order of the small-volume expansion (Lüscher; abstract, B78) and are not proved here. The statement that follows from them is that the exact lattice inequality carries the $g^{2/3}$ scaling of the small-volume corner on its upper side; a lower bound with that scaling remains the open weak-coupling target of STATE.

3. Infinite volume: a necessary condition for the mass gap

Suppose the lattice theory at fixed a and g has infinite-volume gap $\Delta_a = \liminf_{N_s \rightarrow \infty} \Delta_{a,L} > 0$ (T2'), and write $mc^2 = \Delta_a$. Theorem 1 gives, for every L ,

$$\text{Var}_0 \bar{P}_i \leq \frac{\hbar c g^2 N}{4L \Delta_{a,L}}.$$

By translation invariance in the transverse plane, $\text{Var}_0 \bar{P}_i = N_s^{-2} \sum_y C_i(y)$ with $C_i(y) = \langle \text{Re } P_\lambda(x) \text{Re } P_\lambda(x+y) \rangle_c$ the connected two-point function of the spatial Polyakov loop at transverse separation y . Hence

Corollary 2. If the infinite-volume gap is $mc^2 > 0$, then along any sequence of boxes on which $\Delta_{a,L} \geq mc^2/2$,

$$\sum_{y \in \text{transverse plane}} a^2 C_i(y) \leq \frac{\hbar c g^2 N}{2mc^2} L,$$

and the averaged spatial Polyakov loop self-averages at least like $1/L$.

Reading: in a confining phase the spatial Polyakov loop of length L has connected correlations decaying on a transverse scale ξ , and the left side is of order $\xi^2 \langle P^2 \rangle_c$, so the corollary is satisfied with room; in a phase where the loop orders or its correlations do not decay, the left side grows like L^2 and the corollary forbids a gap. It is a necessary condition only, but it is one written entirely in terms of a single gauge-invariant correlation and the coupling, with no reference to the spectrum.

4. Strong coupling: the bound is empty

At $g = \infty$ the links are independent Haar variables, distinct lines are independent, and $\langle |\text{tr } U_1 \cdots U_{N_s}|^2 \rangle = 1$ for a product of independent Haar elements of $SU(N)$ (the product is again Haar distributed), so $\text{Var } P_\lambda = \frac{1}{2}$ for $N \geq 3$ and $\text{Var } \bar{P}_i = 1/(2N_s^2)$. Theorem 1 then reads $\Delta \leq \hbar c g^2 N N_s^2 / (2L)$, which grows with the lattice and says nothing. The averaged loop is the wrong trial operator in the disordered phase, as the plaquette is in the ordered one; the two bounds are complementary, and neither is uniform in a along the scaling curve.

5. Consequence for STATE

The upper side of the small-volume corner is now an exact lattice inequality plus two zero-mode expectation values, and a mass gap has a necessary condition in the Polyakov-loop susceptibility (Corollary 2). The lower side, a bound $\Delta_{a,L} \geq c g^{2/3} \hbar c / L$ at weak coupling in a small box, remains the open weak-coupling target; its natural form, suggested by the shape of Theorem 1, is an operator inequality in which the constant-mode zero-point energy of C133 survives the coupling to the nonzero modes.